

Lesson 7: Poisson distribution

TB sections 3.4

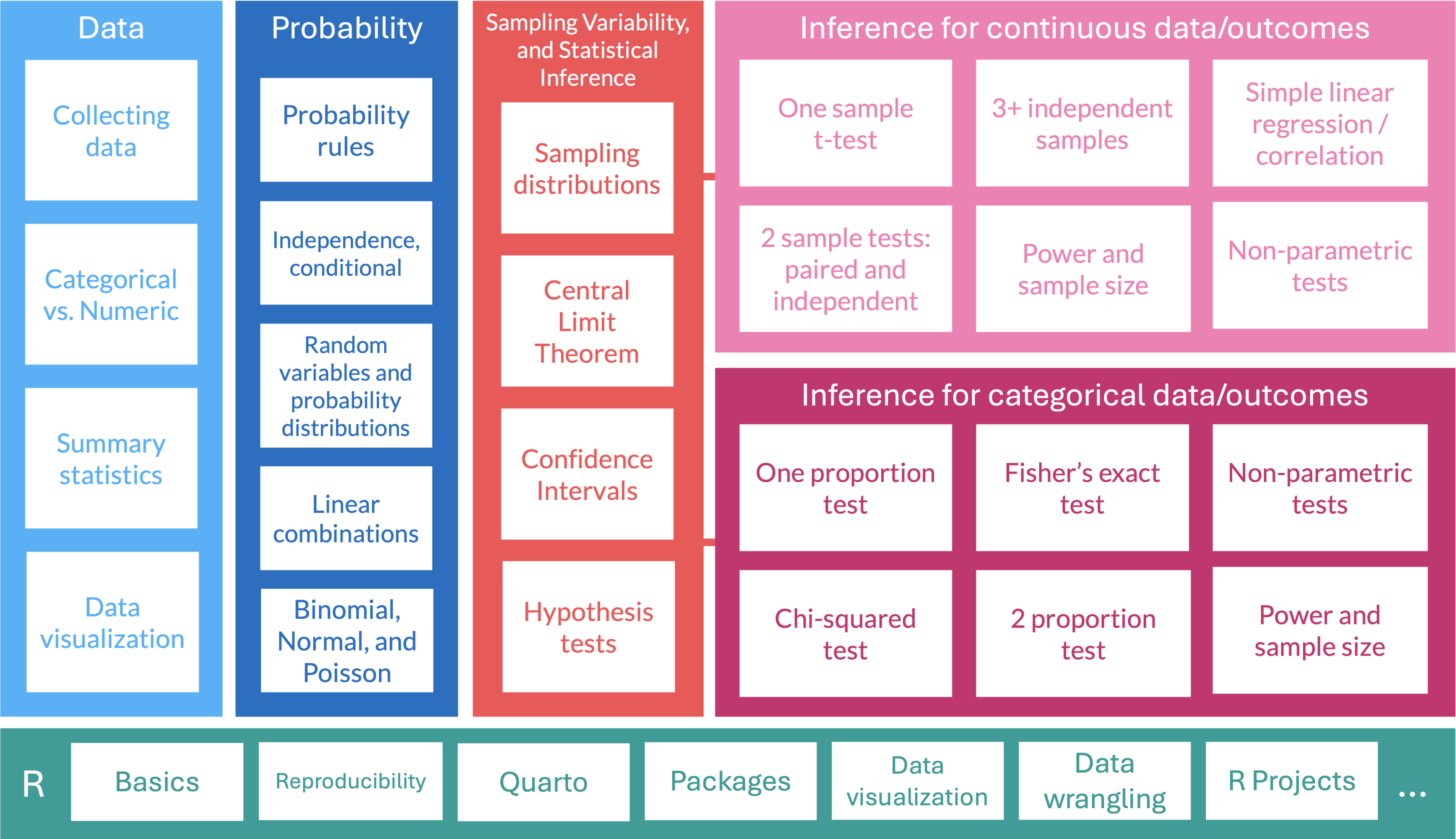
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Learning Objectives

1. Calculate probabilities for different events using a Poisson distribution

Where are we?



Introduction to the Poisson distribution

- The Poisson distribution is often used to model **count data (# of successes)**, especially for **rare events**
 - It is a **discrete distribution**!
- It is used most often in settings where **events happen at a rate λ** per unit of population and per unit time
- **Example:** historical records of hospitalizations in New York City indicate that an average of 4.4 people are hospitalized each day for an acute myocardial infarction (AMI)
 - We can plot the distribution of hospitalizations on each day

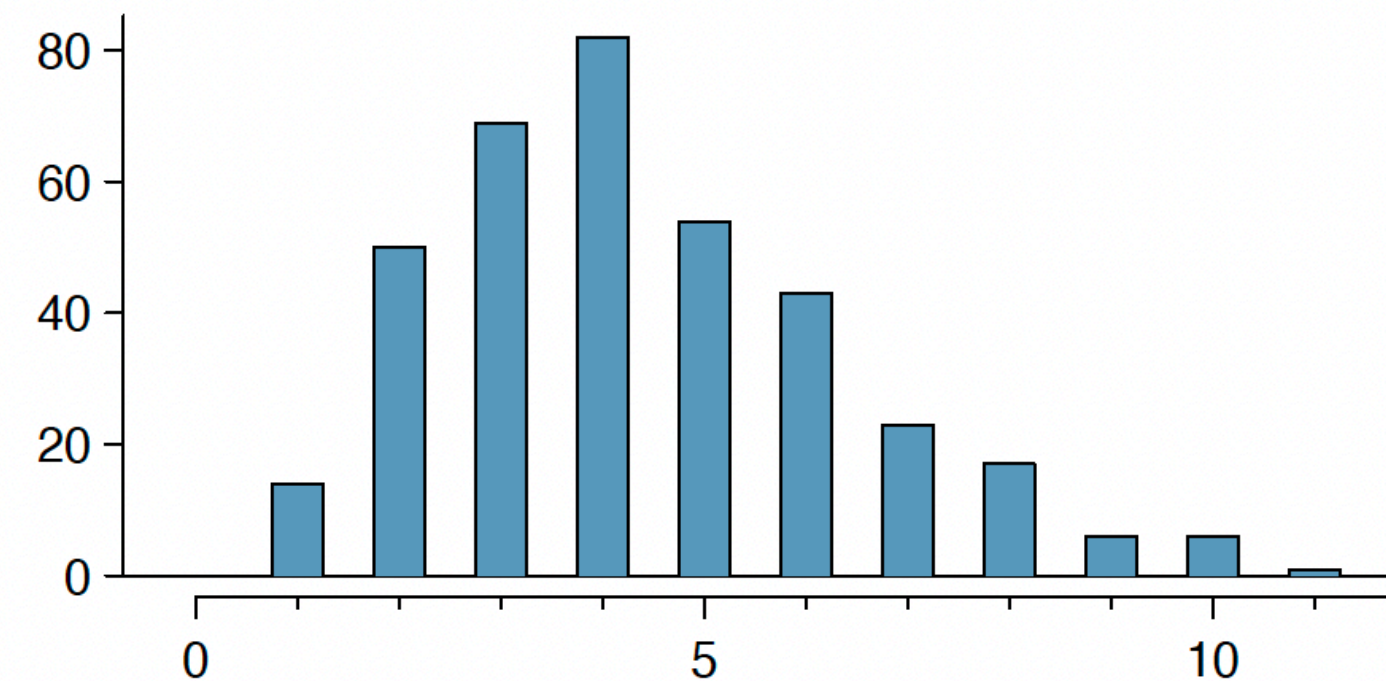


Figure 3.20: A histogram of the number of people hospitalized for an AMI on 365 days for NYC, as simulated from a Poisson distribution with mean 4.4.

Poisson distribution

- Suppose events occur over time in such a way that
 1. The probability an event occurs in an interval is proportional to the length of the interval.
 2. Events occur independently at a rate λ per unit of time.
- Then the probability of exactly x events in one unit of time is

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

- For the Poisson distribution modeling the number of events in one unit of time:
 - The mean is λ .
 - The standard deviation is $\sqrt{\lambda}$.
- Shorthand for a random variable, X , that has a Poisson distribution:

$$X \sim \text{Pois}(\lambda)$$

Poisson distribution: R commands

R commands with their **input** and **output**:

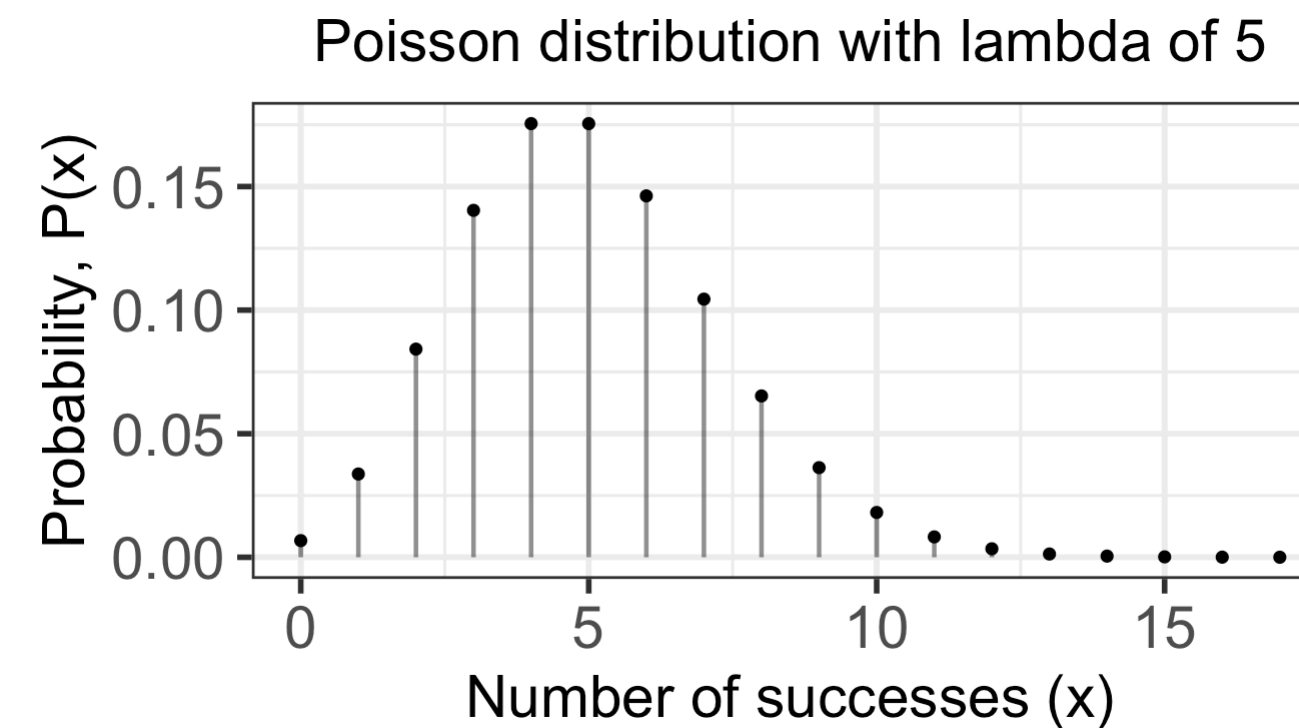
R code	What does it return?
<code>rpois()</code>	returns sample of random variables with specified Poisson distribution
<code>dpois()</code>	returns value of probability density at certain point of the Poisson distribution
<code>ppois()</code>	returns cumulative probability of getting certain point (or less) of the Poisson distribution
<code>qpois()</code>	returns number of cases corresponding to desired quantile

Example: probabilities from a Poisson distribution (1/4)

Typhoid fever

Suppose there are on average 5 deaths per year from typhoid fever over a 1-year period.

1. What is the probability of 3 deaths in a year?
2. What is the probability of 2 deaths in 0.5 years?
3. What is the probability of more than 12 deaths in 2 years?



Example: probabilities from a Poisson distribution (2/4)

Typhoid fever

Suppose there are on average 5 deaths per year from typhoid fever over a 1-year period.

1. What is the probability of 3 deaths in a year?

- $\lambda = 5$ and we want $P(X = 3)$

$$P(X = 3) = \frac{e^{-5}5^3}{3!} = 0.1404$$

```
1 dpois(x = 3, lambda = 5)
[1] 0.1403739
```


Example: probabilities from a Poisson distribution (3/4)

Typhoid fever

Suppose there are on average 5 deaths per year from typhoid fever over a 1-year period.

2. What is the probability of 2 deaths in 0.5 years?

$\lambda = ?$ and we want $P(X = 2)$

- $\lambda = 5$ was the rate for one year. When we want the rate for half year, we need to calculate a new λ :

$$\lambda = \frac{5 \text{ deaths}}{1 \text{ year}} \cdot \frac{1 \text{ year}}{2 \text{ half-years}} = \frac{2.5 \text{ deaths}}{1 \text{ half-year}}$$

$$P(X = 2) = \frac{e^{-2.5} 2.5^2}{2!} = 0.02565$$

```
1 dpois(x = 2, lambda = 2.5)
```

```
[1] 0.2565156
```

Example: probabilities from a Poisson distribution (4/4)

Typhoid fever

Suppose there are on average 5 deaths per year from typhoid fever over a 1-year period.

3. What is the probability of more than 12 deaths in 2 years?

$\lambda = ?$ and we want $P(X > 12)$

- Need to calculate a new λ for 2 years: $\lambda = \frac{5 \text{ deaths}}{1 \text{ year}} \cdot \frac{2 \text{ years}}{1 \text{ two-years}} = \frac{10 \text{ deaths}}{1 \text{ two-year}} = \frac{10 \text{ deaths}}{2 \text{ years}}$

$$P(X > 12) = 1 - P(X \leq 12) = 1 - \sum_{k=0}^{12} \frac{e^{-10} 10^k}{k!} = 0.2084$$

```
1 1 - ppois(q = 12, lambda = 10)
[1] 0.2084435
```

```
1 ppois(q = 12, lambda = 10,
2       lower.tail = F)
[1] 0.2084435
```

Poisson approximation of binomial distribution

- Poisson distribution can be used to approximate binomial distribution when n is large and p is small
 - When Normal approximation does not work
- Binomial distributions for different n (columns) and p (rows)

